Addendum — the face/scene difference is statistically significant

Computed 4 September 2026 by `emoviz/significance.py`. This closes the one inferential gap in the analysis: every subset was previously reported against its own random-pairing null, but the face and scene rates had never been tested **against each other**.

Results

	face-classified	scene-classified	difference
Valence-sign agreement	73.9% [68.2, 79.2]	42.9% [33.3, 51.1]	**+30.9% [+22.6, +40.6]**
Emotion-family agreement	61.9% [55.7, 68.0]	35.6% [26.7, 44.1]	**+26.3% [+18.4, +35.3]**

95% confidence intervals from a clustered bootstrap, 10,000 draws. Clustered permutation test, 10,000 draws: **p = 0.0001** for both metrics.

`inference.py` (9 September), which resamples each subset within itself rather than splitting inside each draw, gives [68.3, 79.2] and [33.1, 51.1] for valence sign and [55.8, 68.1] and [26.4, 44.1] for family — the same intervals to within 0.3 points. Cohen's kappa for family agreement: face 0.30 [0.23, 0.37], scene 0.13 [0.06, 0.19].

Why the resampling is clustered

249 of the 1,012 analysed records repeat an earlier prompt, so records are **not independent**. Both procedures therefore resample and shuffle whole **prompt groups**, never individual records. Resampling records would have understated the uncertainty and inflated significance.

What this settles, and what it does not

Settles: the gap between face-classified and scene-classified drawings is not sampling noise. Neither confidence interval for the difference comes close to zero.

Does not settle: the partial circularity. The face/scene split is derived from the same BLIP-VQA output whose description the image reader then reads. A significant difference shows the two groups differ; it does not show that faces inherently carry more emotional signal. That distinction must be stated whenever the result is quoted.

Note also what actually drives the split: 414 of the 421 scene-classified records (98.3%) are ones where the describer reported **no people present** — and 406 of those *do* contain an expression word. The split is governed by the people-count answer, not by absent expression content.

The sentence to use

> The difference is statistically significant — a clustered permutation test gives p = 0.0001, and the 95% confidence interval on the difference is +22.6 to +40.6 percentage points for valence-sign agreement. I clustered on prompt identity because 249 records repeat an earlier prompt. What that establishes is that the gap is not noise; it does not remove the circularity, because the split comes from the same describer output the emotion reader reads.

Replaces

Wherever the documents or the deck said *"no formal significance test of the face–scene difference has been run"*, that is superseded. The deck was presented on 4 September with the spoken correction; the project documents have since been updated (9 September). The circularity caveat still stands.